

Polynomial Factoring

Factor $x^3 + 1$ and find all roots

Step 1: Use sum of cubes formula

- $x^3 + 1 = (x + 1)(x^2 - x + 1)$
- First root: $x = -1$

Step 2: Quadratic formula on $x^2 - x + 1 = 0$

- $x = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm \sqrt{-3}}{2}$
- $= \frac{1 \pm i\sqrt{3}}{2}$

Step 3: All three roots

- Root 1: $x = -1$
- Root 2: $x = \frac{1+i\sqrt{3}}{2} = e^{i\pi/3}$
- Root 3: $x = \frac{1-i\sqrt{3}}{2} = e^{-i\pi/3}$
- These are the three cube roots of -1

Aliasing

Find alias: $f_s = 44100$ Hz, signal = 32100 Hz

Step 1: Check Nyquist

- Nyquist freq = $f_s/2 = 22050$ Hz
- $32100 > 22050$, so aliasing occurs

Step 2: Calculate alias frequency

- Use: $f_{alias} = |f - n f_s|$ where n minimizes
- $n = 0$: $|32100 - 0| = 32100$ Hz
- $n = 1$: $|32100 - 44100| = 12000$ Hz (smaller!)
- $n = 2$: $|32100 - 88200| = 56100$ Hz (larger)

Step 3: Verify and conclude

- 12000 Hz is in valid range $[0, 22050]$
- The 32100 Hz signal appears as 12000 Hz

DFT Calculation

Find $X[1]$ for $x = [1, 0, 1]$, $N = 3$

Step 1: Write DFT formula

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-i2\pi kn/N}$$

Step 2: Substitute values ($k = 1$, $N = 3$)

- $X[1] = x[0]e^0 + x[1]e^{-i2\pi/3} + x[2]e^{-i4\pi/3}$
- $= 1(1) + 0(e^{-i2\pi/3}) + 1(e^{-i4\pi/3})$
- $= 1 + e^{-i4\pi/3}$

Step 3: Simplify the exponential

- $e^{-i4\pi/3} = \cos(-4\pi/3) + i\sin(-4\pi/3)$
- $= \cos(4\pi/3) - i\sin(4\pi/3)$
- $= -1/2 - i \cdot (-\sqrt{3}/2) = -1/2 + i\sqrt{3}/2$

Step 4: Final answer

- $X[1] = 1 + (-1/2 + i\sqrt{3}/2) = 1/2 + i\sqrt{3}/2$
- This equals $e^{i\pi/3}$
- DFT coefficient: $c_1 = X[1]/N =$

FIR Filter Analysis

Analyze $y_t = x_t - (1/4)x_{t-1}$

Step 1: Transfer function

- Direct: $H(z) = 1 - (1/4)z^{-1}$

Step 2: Find zeros and poles

- Zero: Set $1 - (1/4)z^{-1} = 0$
- Solve: $z^{-1} = 4$, so $z = 1/4$
- Pole: At $z = 0$ (from z^{-1} term)
- Zero at $z = 1/4$, pole at $z = 0$

Step 3: Frequency response at $\omega = 0$

- $H(0) = 1 - (1/4)e^{-i \cdot 0} = 1 - 1/4 = 3/4$
- DC gain is 0.75 (attenuation)

Step 4: Magnitude at $\omega = \pi/2$

- $H(\pi/2) = 1 - (1/4)e^{-i\pi/2}$
- $e^{-i\pi/2} = -i$, so $H(\pi/2) = 1 - (1/4)(-i) = 1 + i/4$
- $|H(\pi/2)| = |1 + i/4| = \sqrt{1 + 1/16} =$

IIR Filter with Impulse

Analyze $y_t = x_t - (1/4)y_{t-1}$, impulse response

Step 1: Find transfer function

- Rearrange: $y_t + (1/4)y_{t-1} = x_t$
- Z-transform: $Y(z)[1 + (1/4)z^{-1}] = X(z)$
- $H(z) = 1/[1 + (1/4)z^{-1}] = z/(z + 1/4)$

Step 2: Check stability

- Pole at $z = -1/4$
- $|pole| = |-1/4| = 1/4 < 1$ STABLE

Step 3: Impulse response calculation

- Input: $x = [1, 0, 0, \dots]$

- $y_0 = 1 - (1/4)(0) = 1$
- $y_1 = 0 - (1/4)(1) = -1/4$
- $y_2 = 0 - (1/4)(-1/4) = 1/16$
- $y_3 = 0 - (1/4)(1/16) = -1/64$

Step 4: Pattern recognition

- $h[n] = (-1/4)^n$ for $n \geq 0$
- Alternating sign, exponential decay
- Sum: $\sum h[n] = 1/(1 - (-1/4)) = 4/5$

Reson Filter Design

Design reson: $\omega = \pi/4$, bandwidth $B = 0.1$

Step 1: Calculate pole radius from bandwidth

- Narrow band formula: $B \approx 2(1 - R)$
- $0.1 = 2(1 - R)$, so $1 - R = 0.05$
- $R = 0.95$

Step 2: Place complex conjugate poles

- Poles at $z = 0.95e^{\pm i\pi/4}$
- In rectangular: $0.95 \cdot (\frac{1}{\sqrt{2}} \pm i\frac{1}{\sqrt{2}})$
- $= \frac{0.95}{\sqrt{2}}(1 \pm i) = 0.672(1 \pm i)$

Step 3: Write transfer function

- $H(z) = \frac{1}{1 - 2R\cos(\pi/4)z^{-1} + R^2z^{-2}}$
- $\cos(\pi/4) = 1/\sqrt{2}$, so $2R\cos(\pi/4) = 2(0.95)/\sqrt{2} = 1.343$
- $H(z) = \frac{1}{1 - 1.343z^{-1} + 0.9025z^{-2}}$

Step 4: Calculate peak gain

- At resonance ($\omega = \pi/4$):
- Approximate gain $\approx 1/(1 - R) = 1/0.05 = 20$
- In dB: $20\log_{10}(20) = 26$ dB

Plucked String Tuning

Tune to 440 Hz, $f_s = 44100$ Hz

Step 1: Calculate ideal period in samples

- $P = f_s/f_0 = 44100/440 = 100.227$ samples
- Need to achieve this total delay

Step 2: Allocate delays

- Try integer delay $L = 100$
- Low-pass filter adds 0.5 samples
- Total so far: $100 + 0.5 = 100.5$

- Still need: $100.227 - 100.5 = -0.273$ (negative!)

Step 3: Adjust allocation

- Use $L = 99$ instead
- With LP: $99 + 0.5 = 99.5$

- Need from all-pass: $100.227 - 99.5 = 0.727$

Step 4: Calculate all-pass coefficient

- For delay $\delta = 0.727$:
- $a = (1 - \delta)/(1 + \delta) = 0.273/1.727 = 0.158$

Step 5: Verify total delay

- Integer: 99 samples
- Low-pass: 0.5 samples
- All-pass: 0.727 samples

- Total: $99 + 0.5 + 0.727 = 100.227$ Perfect!

All-Pass Identification

Is $H(z) = (0.5z + 1)/(z + 0.5)$ all-pass?

Step 1: Convert to standard form

- Divide num and denom by z :
- $H(z) = (0.5 + z^{-1})/(1 + 0.5z^{-1})$

Step 2: Check all-pass structure

- All-pass form: $H(z) = (a + z^{-1})/(1 + az^{-1})$
- Our filter: numerator has $a = 0.5$
- Denominator also has $a = 0.5$
- Structure matches! YES, all-pass

Step 3: Verify magnitude = 1

- At any frequency: $|H(e^{i\omega})| = |(0.5 + e^{-i\omega})/(1 + 0.5e^{-i\omega})|$
- Numerator and denominator have same structure
- Can show $|H(e^{i\omega})| = 1$ for all ω

Step 4: Phase delay property

- All-pass changes phase without changing magnitude
- Used for fractional delay adjustment
- Phase: $\phi(\omega) = -2\arctan(\frac{0.5\sin\omega}{1+0.5\cos\omega})$